

Microcontroller Reflection

Our goal for the microcontroller section was to use a microcontroller and breadboard to program a circuit to complete various tasks. We began this journey by learning how to tackle programming problems. We were taught general programming approaches on how to solve problems and the importance of in-depth planning and documentation. We learned how to construct visual plans like flow charts with symbols to understand how to solve programming problems. We read about Dr. Valvano's security system design and requirements and identified different variables, like inputs and outputs and constraints. We applied our knowledge to create system diagrams and flowcharts of what the system would look like and practiced making other flow charts. After that we created Github accounts to upload our coding projects to our portfolios.

We received our Nano microcontrollers and learned about the different pins and what their functions are. After that we were introduced to micropython coding and how to execute certain tasks. Next class, we learned about circuits and how to read schematics. We learned how to put various pieces into a breadboard and how to connect them to each other or ground/power. We build our own circuit on the breadboard and applied more voltage and the resistor burnt out. We were taught about the importance of fail safes and how they usually take the form of resistors. After that we swapped the switch on our breadboard for a light sensor. We used the light sensor to redesign Dr. Valvano's security system with different parts. We made flow charts and systems designs to plan before making our circuit. My group used the light sensor connected with the green light to show when the device was armed at night. We used the button with the yellow light to represent the touch sensor and the activation of the alarm in the event of an intruder. We used AI to write our code for the microcontroller and successfully hooked up our circuit to execute its functions. We did have to change our pins and tweak the code but we got our circuit to run successfully.

Next class we created an embedded system with the goal of rotating the servo arm 180 degrees. We used our microcontroller and AI to create a code to execute the rotation. We then tested a rain drop sensor and how it performs under different conditions using a digital multimeter. We conducted different tests on different conditions on the sensor to see how the VDC varied. After doing this manually we connected it to our arduino and had AI write code to collect 10 samples in a csv file. We conducted the same tests on the automated system. We created a graph with the averages and standard deviation with lines of best fit. After much trial and error we completed the tasks and uploaded everything to our Github accounts.

In this section we learned about microcontrollers in embedded systems and how they are coded to execute different functions. We learned how to wire them on a breadboard connected to different sensors and how to collect data from it.